



CMT 307 Coursework 2 Group Project

Group Number	9
Project Title	Hate Speech Detection
Supervisor	Ogechukwu Ukwandu
Word Count	2,704
Student IDs	22119242 23093280 23084873 23060968 23114859 23108373 23090089 2060124

Table of Contents

1. Introduction.....	3
2. Literature Review.....	4
3. Dataset.....	5
3.1. Tweet Analysis.....	5
3.1.1. Tweet Length (Word Count).....	5
3.1.2. Frequent Words.....	6
4. Methodology.....	7
4.1. Preprocessing.....	7
4.2. Proposed Models.....	8
5. Experimental Setting.....	8
6. Results.....	9
7. Analysis.....	10
8. Conclusion.....	10
9. Future Work.....	11

1. Introduction

In the current era of social computing, media platforms and chat forums, microblogging has enabled people to express their views, debate and defend their opinions and argumentations, enabling exponential growth and revolutionising social communication instantaneously and extensively [1]. Social media platforms offer anonymity, easy access, and rapid communication, which has led to the dissemination of hate speech, leading to an increase in cyber-hate crimes in recent years [1]. Determining whether a segment of text includes hate speech is a challenging task. Hate speech represents a multifaceted occurrence that is inherently intertwined with intergroup dynamics and dependent on the subtleties of language [2].

Whilst Hate speech lacks a universal definition, there is a consensus that it is a speech which targets disadvantaged social groups in a manner that is potentially harmful to them[1]. The International Covenant on Civil and Political Rights (ICCPR) defines *hate speech* as “any advocacy of national, racial or religious hatred that constitutes the incitement to discrimination, hostility or violence” [3]. According to Twitter’s hateful conduct policy, *hateful conduct* can be defined as an attack on “other people on the basis of race, ethnicity, national origin, caste, sexual orientation, gender, gender identity, religious affiliation, age, disability, or serious disease.”[8]

The Polish Ministry of Science and Higher Education funded & published a study which showed that exposure to hate speech has adverse effects on people who view it, irrespective of whether they are part of the intended group or not [9]. More importantly, hate speech has been found to cause a multitude of psychological and emotional harm to the recipient of the hate speech. Additionally, at a larger scale, hate speech can prolong the entrenchment of prejudice and worsen racial divisions by dehumanising targets and instilling an “us” vs “them” mentality [9].

For these reasons, detecting abusive language and hate speech has increased interest recently. Google created “Perspective” to determine toxic comments online based on training data in the form of millions of annotated (very toxic to very healthy) comments [10]. Additionally, an abuse-detecting algorithm with 90% accuracy using a combination of machine learning and crowdsourcing has been developed at Yahoo [11]

This work focuses on Twitter, a social media platform that allows users to post tweets. The objective is to explore machine learning techniques and natural language processing to identify hate speech content in the Twitter dataset. Different preselected feature generation, resampling, and classification algorithms were applied to the dataset to develop a decision system that detects tweets that contain elements of hate speech.

2. Literature Review

A diverse range of machine learning algorithms and techniques have been proposed for effective hate speech detection. A chronological order is given below [12].

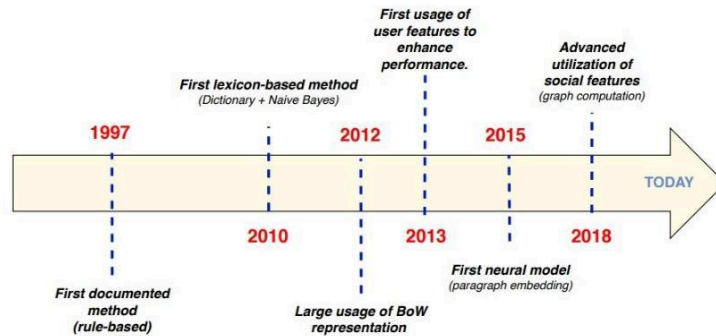


Figure 1. Milestones in the field of Hate Speech Detection in chronological order.

SemEval-2019 Task 5 had 74 teams that participated, of which there were 69 submissions for English Sub-task A that handles Hate Speech Detection Against Immigrants and Women. Team YNU_DYX employed a stacked Bidirectional Gated Recurrent Units (BiGRUs) with 0.55 targeted dropout and 64 units per layer. This yielded an accuracy score of 0.56, precision of 0.64, recall of 0.60, and an f1 score of 0.55 [4]

Team ABARUAH employed Bidirectional LSTM with GloVe word embeddings as a preprocessing method. The dropout rate was set to 0.25 and the number of units per layer is set to 100. The model yielded an accuracy score of 0.748, precision of 0.749, recall of 0.748, and an f1 score of 0.748 [5]

Attention based LSTM-RNN system was proposed by [6] at SemEval 2019 for Multilingual detection of hate speech against immigrants and women on Twitter (HatEval-Task 5). It consisted of two subtasks namely subtask A (hate-speech detection against immigrants and women) and subtask B (aggressive behaviour and target classification). For the english tweets, they achieved an accuracy of 0.653, F1 score of 0.651, precision of 0.69 and recall of 0.679 for subtask A. For subtask B they achieved a F1 score of 0.467 and EMR of 0.57.

A machine learning strategy based on implementing a CNN using pre-trained word embeddings (GloVe and FastText) with 0.5 dropout was proposed by [7] at SemEval-2019 Task 5. They achieved an accuracy of 0.520, F1 score of 0.488, precision of 0.628 and recall of 0.574 using CNN-FastText on English Twitter tweets subtask A (hate-speech detection against immigrants and women). For sub-task B (aggressive behaviour and target classification) they achieved a F1 score of 0.577 and EMR of 0.297.

3. Dataset

The following section discusses the dataset that has been used in this project. The dataset that has been provided is the SemEval's HatEval english tweets dataset. The dataset consists of 9000 tweets on the train data set. Of the provided tweets 42% are classed as hate speech tweets while 58% are not.

Intention	Filename	Rows	Columns
Training	train_en.csv	9000	5
Validation	dev_en.csv	1000	5
Testing	test_en.csv	2970	5

Table 1: Shape of data files available

Each dataset has a similar structure, with the same columns: id, text, HS, TR, AG. For this task, the focus will only be on predicting if a tweet is classified as Hate Speech (0: Absent | 1: Present), and only keep the "text" and "HS" columns.

	id	text	HS	TR	AG
0	201	Hurray, saving us \$\$\$ in so many ways @potus @...	1	0	0
1	202	Why would young fighting age men be the vast m...	1	0	0
2	203	@KamalaHarris Illegals Dump their Kids at the ...	1	0	0

Fig1: First Three Rows of the Dataset

The Hate Speech distribution for the datasets is given below:

Column	Dataset	Label (0)	Label (1)
Hate Speech	Train	5,217	3,783
	Dev	573	427
	Test	1,719	1,252

Table 2: Dataset label statistic

3.1. Tweet Analysis

It was important to carry out an analysis of the characteristics of tweets text in the available corpora. As the structure of the datasets is similar, we carry out this analysis on the training dataset.

3.1.1. Tweet Length (Word Count)

As seen in the following plots, tweets with hate speech are slightly longer on average than those without hate speech, whereas in both cases, the deviation about the mean is similar. At the same time, most quartiles show that tweets with hate speech are generally slightly longer than those labelled as not hate speech.

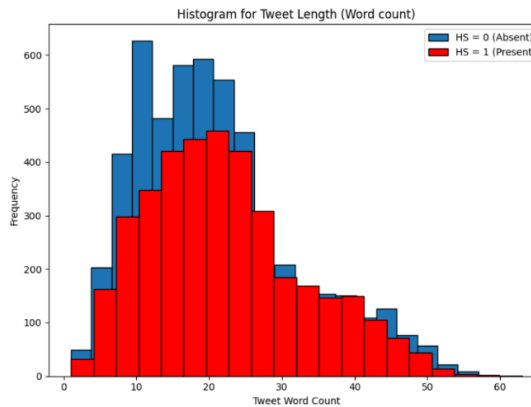


Figure 2: First Three Rows of the Dataset

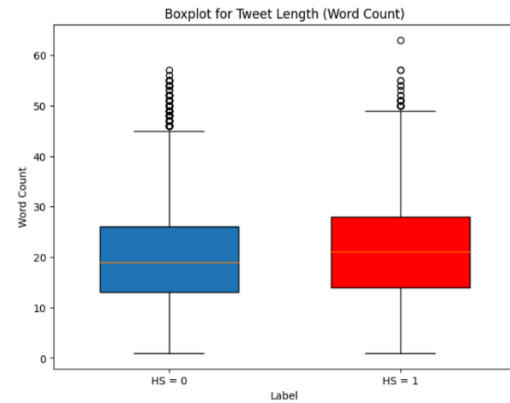


Fig3: First Three Rows of the Dataset

Statistic	Value (No HS)	Value (HS)
Count	5217	3783
Mean	20.72	22.18
Standard Deviation	10.96	10.64
Minimum	1	1
Q1	13	14
Median	19	21
Q3	26	28
Maximum	57	63

Table 3: Tweet Length (Word Count) Distribution

3.1.2. Frequent Words

After removal of URLs, twitter hashtags and handles and words with single characters, we see a clear distinction between tweets belonging to each of the hate speech labels, although not without some offensive words in tweets labelled as HS = 0. We can also see the presence of a few stop words.

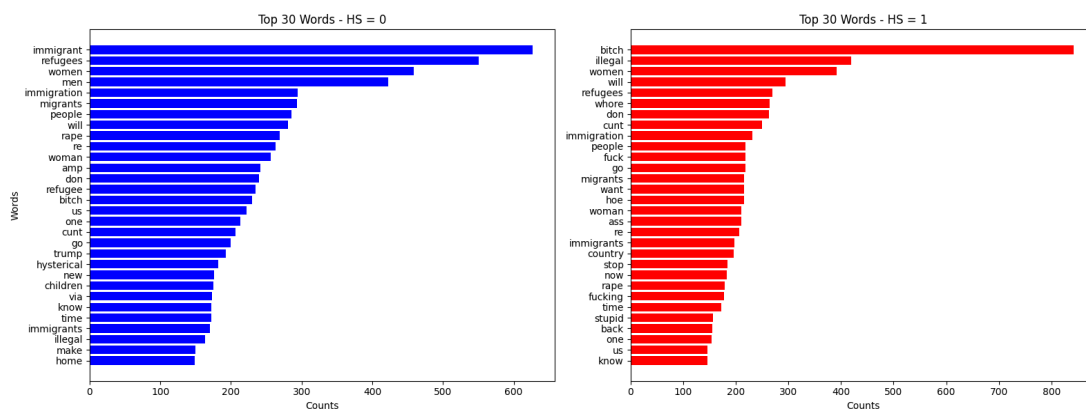


Figure 4: Frequent words distribution

4. Methodology

4.1. Preprocessing

For preprocessing, given a tweet, we applied some popular techniques to standardise the text:

- Lowercasing.
- Normalise the top 50 (and related) hashtags present in the hate speech training set into words, for example “#maga” becomes “make america great again”. The other hashtags have been removed.

An extract of the top hashtags align with their frequency and normalisation is shown below:

<i>Hashtags</i>	<i>Frequency</i>	<i>Normalisation</i>
#buildthatwall	317	build that wall
#maga	196	make america great again
#buildthewall	166	build the wall
#womensuck	119	women suck
#nodaca	118	no daca

Table 4: Hashtag, Frequency and Normalisation

- Tokenizing (using nltk’s TweetTokenizer).
- Remove short words (of length of 1 and 2) and stop words except for “no” and “not”, since the omission of them can change the connotation of the tweet.
- Stemming using nltk’s SnowballStemmer.
- Remove links, usernames, numbers and special characters.

An example of how the preprocessing transforms the tweet is exemplified below (the username is anonymised).

Original Tweet	Now his slag mother has been blown up hopefully that lil cunt gets it nxt as well https://t.co/BTmxENcXKR
Cleaned Tweet	slag mother blown hope lil cunt get nxt well

Once the preprocessing was done, we created a text vectorization layer, computing a vocabulary of 5,002 words adapted from the training set and limited the number of tokens per tweet to 25. So if the tweet had fewer than 25 (number of tokens), it would be padded and truncated if it exceeded. We varied vocabulary size (10,002, 50,002 and 100,002 words), but increasing it too much led to the

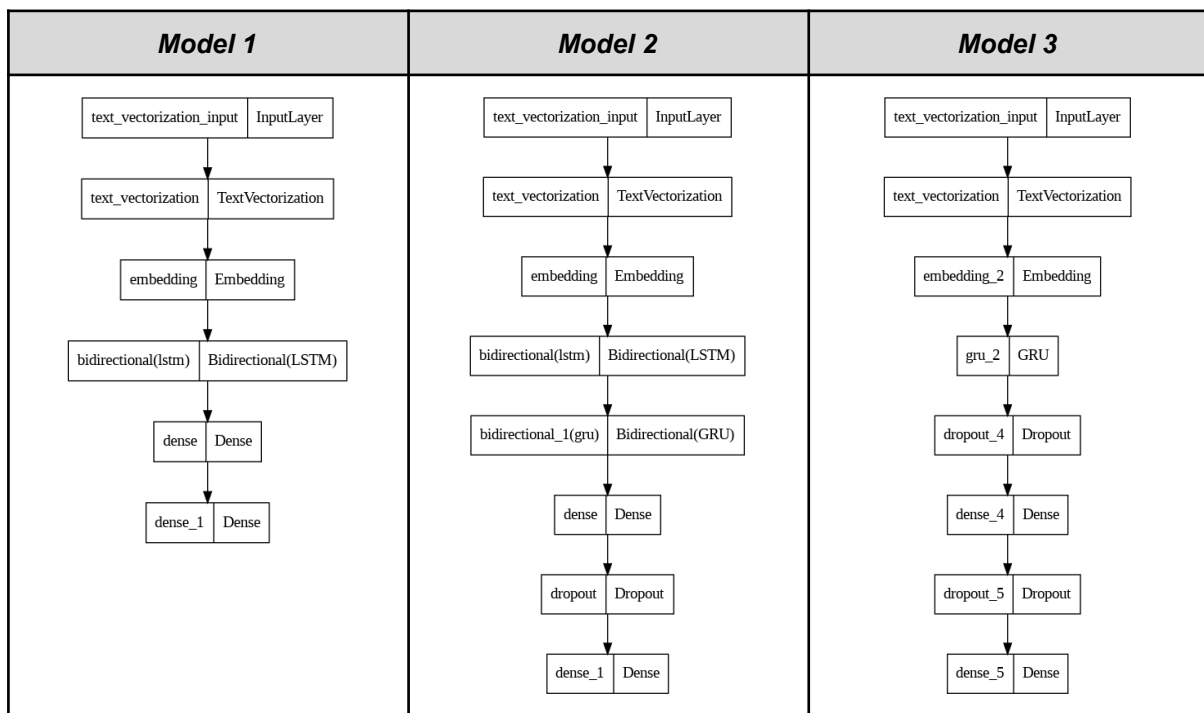
model showing clear signs of overfitting with few epochs (3 - 5), such as high accuracy in the training set but showed an increasing loss in the validation set.

4.2. Proposed Models

We worked with three RNN models:

- Model 1: Bidirectional LSTM (Bi-LSTM)
- Model 2: Bidirectional LSTM-GRU (Bi-LSTM-GRU)
- Model 3: GRU (GRU)

The following plots show the architecture for each model.



5. Experimental Setting

Our datasets were separated into groups of size 64 (batch size), and our training set was also shuffled to ensure the randomness and uniformity of the groups.

Initially our models were trained with manually set parameters for ten epochs for the information regarding accuracy, precision, recall and F1 score, which would then serve as our baseline. For all models the output dimension of the embedding layer was 64. The first model had 64 units on the Bi-LSTM layer and a learning rate of 0.0001, the second model had 64 and 32 units in the Bi-LSTM and Bi-GRU layers respectively, a 0.2 dropout rate in the Bi-GRU layer and a learning rate of 0.001, and the third model had 128 units in the GRU layer, 64 units on the first densely connected layer, a dropout rate of 0.5 in both dropout layers and a learning rate of 0.001.

To improve the performance of the models, we have used Keras' Hyperband Tuner, which speeds up random search through adaptive resource allocation and early-stopping, to get the optimal parameters that yielded the best accuracy in the validation set.

For the Bi-LSTM model we'll get the optimal number of units in the Bidirectional LSTM layer. Similarly for the Bi-LSTM-GRU, we'll also get the optimal number of units for the Bidirectional LSTM layer (also related to the number of units for the Bidirectional GRU layer) and the optimal dropout rate for the GRU layer. For the GRU model we'll get the optimal number of units in the GRU layer (also related to the units in the first densely connected layer) and the dropout rate for the first Dropout layer. Additionally we'll also get the optimal learning rate for the Adam optimiser for each model.

<i>Model</i>	<i>Units</i>	<i>Dropout</i>	<i>Learning Rate</i>
Bi-LSTM	320	NA	0.001
Bi-LSTM-GRU	128	0.5	0.001
GRU	256	0.2	0.001

6. Results

Our models have been trained for 50 epochs to envision the behaviour of the metrics of our tuned model. Our baseline and tuned models showed clear signs of overfitting, such as a continuous increase in the validation loss without any significant improvement in accuracy and F1 score, as the models reached an accuracy higher than 99% on the training set. To overcome this issue, we re-trained the tuned models with their optimal parameters and ran them for five epochs to allow enough training while preventing a significant increase in the validation loss.

The performance of the three re-trained tuned models on the validation and test sets are shown in the following tables:

<i>Validation Set</i>					
<i>Model</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Recall</i>	<i>Precision</i>	<i>F1 Score</i>
Bi-LSTM	0.8717	0.6958	0.6897	0.6356	0.6578
Bi-LSTM-GRU	0.9835	0.6990	0.6509	0.6508	0.6464
GRU	0.8203	0.6865	0.6427	0.6310	0.6333

<i>Test Set</i>					
<i>Model</i>	<i>Loss</i>	<i>Accuracy</i>	<i>Recall</i>	<i>Precision</i>	<i>F1 Score</i>
Bi-LSTM	1.6253	0.5153	0.8764	0.4621	0.6018
Bi-LSTM-GRU	1.6885	0.5227	0.8530	0.4647	0.5987

GRU	1.2940	0.5093	0.8622	0.4588	0.5954
------------	--------	--------	--------	--------	--------

7. Analysis

All the developed models performed similarly on both the validation and test datasets, nevertheless we observed a significant increase in loss in the test dataset from what we saw from the validation loss, as well as nonsimilar metrics, namely recall and precision that are around 0.63 and 0.69 on the validation dataset, whereas in the test dataset the recall is around 0.85 and precision is around 0.46. This could be caused by a mismatch between the datasets, since it seems validation and test sets do not come from the same distribution, which does not allow an appropriate tuning and limits the generalisation power of our models.

This is easily appreciated by looking at the confusion matrices for the three models, where we can see that our model labels most of the tweets as hate speech.

		<i>Model 1</i>		<i>Model 2</i>		<i>Model 3</i>	
		Predicted		Predicted		Predicted	
		0	1	0	1	0	1
Actual Class	0	385	1,334	407	1,312	351	1,368
	1	130	1,122	154	1,098	139	1,113

We also noticed that due to the difference in distribution there were some words present in tweets that the models immediately associated with hate speech while there were instances where that wouldn't necessarily be true. For example, taking the false positives across the 3 models, i.e. all the models classify incorrectly as hate speech, almost half of them (534 of 1,161) contain the word "bitch". Applying some type of regularisation on the network could fix this problem to reduce the importance of tokens that have that effect.

This issue can also be seen in the submissions to the SemEval competition where they mention the errors are highly skewed towards false positives and the inability of the models to infer if a certain word is used in a humorous or offensive context [13].

8. Conclusion

In conclusion, this project has demonstrated the potential of machine learning techniques, specifically Recurrent Neural Networks, in detecting hate speech on Twitter. Preprocessing techniques applied, including lowercasing, tokenizing, stemming, and removal of links, usernames, numbers, and special characters, have proven effective in standardising the text data for analysis. The

three models used in this study, Bidirectional LSTM (Bi-LSTM), Bidirectional LSTM-GRU (Bi-LSTM-GRU), and GRU, have shown promising results in identifying hate speech content. Despite signs of overfitting, the models achieved balanced metrics after retraining with optimal parameters for a limited number of epochs. Possible solutions or improvements to the models utilised are proposed on future work based on what we observed from the performance.

The study contributes to the growing research in hate speech detection, particularly against immigrants and women on Twitter. The findings highlight the importance of continuous efforts in refining machine learning techniques and models to improve the accuracy of hate speech detection, thereby making social media platforms safer and more inclusive. The fight against online hate speech is a complex and ongoing challenge. However, with machine learning and natural language processing advancements, we are better equipped to tackle this issue and foster healthier online interactions.

9. Future Work

For future work, the current models could benefit from a more deep analysis of the hashtags by either cleaning them as we did with the most popular ones and assign them more importance while building a targeted vocabulary (word weighting), in such a way that the included words have a strong relationship with hate speech tweets.

There is also the possibility of exploring other models. We can explore models like Bidirectional Encoder Representations from Transformers (BERT) which is a language model based on the transformer architecture, notable for its dramatic improvement over previous state of the art models. One other option is using RoBERTa, which is implemented in PyTorch and is trained on a much larger dataset and using a more effective training procedure than BERT.

We can also create new features from the text data that may help the model's performance. This could include sentiment scores, readability scores, or other linguistic features like Part of Speech tagging, or Term Frequency-Inverse Document Frequency (TF-IDF) scores to reflect how important a word is to a document in a collection or corpus. N-grams could be used to help capture some context which might be lost when using single words as features. N-grams are contiguous sequences of n items from a given sample of text or speech.

References

- [1] Md Saroar Jahan, Mourad Oussalah "A systematic review of hate speech automatic detection using natural language processing," May.2023, doi:<https://doi.org/10.1016/j.neucom.2023.126232>.
- [2] Fatimah Alkomah and Xiaogang Ma "A Literature Review of Textual Hate Speech Detection Methods and Datasets." May 2022, <https://doi.org/10.3390/info13060273>.
- [3] Kevin Boyle. Hate speech - The United States Versus the Rest of the World. Me. L. Rev., 53:487, 2001. [Online]. Available at: <https://digitalcommons.maine.gov/mlr/vol53/iss2/7>
- [4] Y. Ding, X. Zhou, and X. Zhang, "YNU_DYX at SemEval-2019 Task 5: A Stacked BiGRU Model Based on Capsule Network in Detection of Hate," Jan. 2019, doi: <https://doi.org/10.18653/v1/s19-2096>.
- [5] Arup Baruah, F. A. Barbhuiya, and K. Dey, "ABARUAH at SemEval-2019 Task 5 : Bi-directional LSTM for Hate Speech Detection," Jan. 2019, doi: <https://doi.org/10.18653/v1/s19-2065>.
- [6] L. De, "GL at SemEval-2019 Task 5: Identifying hateful tweets with a deep learning approach.," Jan. 2019, doi: <https://doi.org/10.18653/v1/s19-2073>.
- [7] A. Ribeiro and Felipe, "INF-HatEval at SemEval-2019 Task 5: Convolutional Neural Networks for Hate Speech Detection Against Women and Immigrants on Twitter," Jan. 2019, doi: <https://doi.org/10.18653/v1/s19-2074>.
- [8] Twitter, "Hate-Speech". [Online]. <https://help.twitter.com/en/rules-and-policies/hateful-conduct-policy>
- [9] Soral, Wiktor, Michaffdfdd Bilewicz, and Mikoffgffdj Winiewski. "Exposure to hate speech increases prejudice through desensitization". September 2017, doi: 10.1002/ab.21737.
- [10] Google. What if technology could help improve conversations online?, 2017. <https://www.perspectiveapi.com/#/>.
- [11] Lulu Chang, New Yahoo algorithm can spot online abuse in context, not just content, 2016. <https://finance.yahoo.com/news/yahoo-algorithm-spot-online-abuse-225438302.html>.
- [12] Edoardo Mosca. "Explainability of Hate Speech Detection Models". April 2020. Masters Thesis. Department of Mathematics, Technische Universitat Munchen.
- [13] Basile et al., "SemEval-2019 Task 5: Multilingual Detection of Hate Speech Against Immigrants and Women in Twitter. June 2019 doi: 0.18653/v1/S19-2007.